

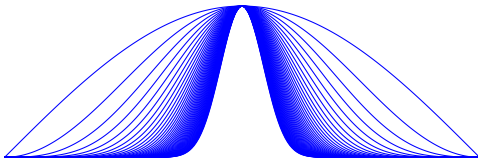
M303 Book D Day School

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This tutorial will be recorded



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Appendix on Mixed Metrics

Question 1

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- (a)
 - (i) Give an example (a_n) of an unbounded strictly increasing monotonic sequence of real numbers with the additional property that $a_n > 4$ for all $n \in \mathbb{N}$.
 - (ii) Give an example (b_n) of a bounded strictly decreasing monotonic sequence of real numbers with the additional property that $b_n < 2$ for all $n \in \mathbb{N}$.
 - (iii) Give an example (c_n) of a convergent sequence of real numbers with the additional property that even terms are greater than 3 and odd terms are less than 3.
- (b) Give an example of a single sequence (x_n) that has each of the three sequences in part (a) as a subsequence.

Brief Answer

- (a) Possible answers: $a_n = 4 + n$, $b_n = 1 + \frac{1}{2n}$, $c_n = 3 + \frac{(-1)^n}{n}$
- (b) Just stitch them together! Like this
 $a_1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3, \dots$, which could be written as:

$$x_{3n-2} = a_n,$$

$$x_{3n-1} = b_n,$$

$$x_{3n} = c_n,$$

for $n \in \mathbb{N}$.

And this may be re-written as follows. For each $n \in \mathbb{N}$, define

$$x_n = \begin{cases} a_{\frac{n+2}{3}}, & \text{when } n \equiv 1 \pmod{3}, \\ b_{\frac{n+1}{3}}, & \text{when } n \equiv 2 \pmod{3}, \\ c_{\frac{n}{3}}, & \text{when } n \equiv 0 \pmod{3}. \end{cases}$$

The Metric Axioms

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A *metric* is a generalisation of Euclidean distance.

Let X be a set. A metric on X is a function $d : X \times X \rightarrow \mathbb{R}$ that satisfies the following three conditions.

- (M1) For all x, y , $d(x, y) \geq 0$ with $d(x, y) = 0$ if, and only if, $x = y$.

The distance between distinct points is positive; the distance from a point to itself is zero.

- (M2) For all x, y , $d(x, y) = d(y, x)$.

The distance between two points is the same, irrespective of 'direction'.

- (M3) For all x, y, z , $d(x, z) \leq d(x, y) + d(y, z)$.

The triangle inequality: the distance between two points x, z cannot become shorter by going via a third point y .

Question 2

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Here are a number of possible distance functions on $\mathbb{R}$. Which, if any, is a metric on $\mathbb{R}$.

(a) $d_1(x, y) = \left| \sqrt{|x|} - \sqrt{|y|} \right|,$

(b) $d_2(x, y) = \left| \sqrt[3]{x} - \sqrt[3]{y} \right|,$

(c) $d_3(x, y) = \max(x, y) - \min(x, y).$

(d) $d_4(x, y) = \min(x, y) - \max(x, y),$

(e) $d_5(x, y) = e^x + e^y,$

(f) $d_6(x, y) = \begin{cases} 0 & \text{if } x = y, \\ e^x + e^y & \text{if } x \neq y. \end{cases}$

Brief Answer

- (a) $d_1(x, y) = \left| \sqrt{|x|} - \sqrt{|y|} \right|$ is not a metric: $d_1(-1, 1) = 0$, so (M1) fails.
- (b) $d_2(x, y) = \left| \sqrt[3]{x} - \sqrt[3]{y} \right|$ is a metric: it is the pull-back of $d^{(1)}$ via the one-one function $f: \mathbb{R} \rightarrow \mathbb{R}$, where $f(x) = \sqrt[3]{x}$ (See [HB, p74](#)).
- (c) $d_3(x, y) = \max(x, y) - \min(x, y)$ is a metric. If you give it some thought you'll see that $\max(x, y) - \min(x, y)$ is just an alternative way of expressing $|x - y|$.
- (d) $d_4(x, y) = \min(x, y) - \max(x, y)$ is not a metric: it fails (M1).
- (e) $d_5(x, y) = e^x + e^y$ is not a metric as $d_5(x, y) > 0$, for all $x, y \in \mathbb{R}$.

Brief Answer

$$(f) \quad d_6(x, y) = \begin{cases} 0 & \text{if } x = y, \\ e^x + e^y & \text{if } x \neq y. \end{cases}$$

This does define a metric; we will check the three conditions.

- (M1) Clearly, by definition, $d_6(a, b) \geq 0$ for all $a, b \in \mathbb{R}$, and $d_6(a, a) = 0$ for all $a \in \mathbb{R}$. Also if $a \neq b$, then line 2 of the definition applies which tells us that $d_6(a, b) > 0$. Thus $d_6(a, b) = 0$ if, and only if, $a = b$.
- (M2) If $a = b$, then $d_6(a, b) = d_6(b, a) (= 0)$.
If $a \neq b$, then $d_6(a, b) = e^a + e^b = e^b + e^a = d_6(b, a)$.
- (M3) We may assume that a, b, c are all distinct. (Why?) In this situation, we have
$$d_6(a, c) = e^a + e^c \leq e^a + 2e^b + e^c = (e^a + e^b) + (e^b + e^c) = d_6(a, b) + d_6(b, c).$$

Having checked all three conditions we conclude that d_6 is a metric on $\mathbb{R}$.

Question 3

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The expression $d(x, y) = |\log x - \log y|$ defines a distance function on the interval $(0, \infty)$. Is it a metric?

Brief Answer

(M1) Clearly, by definition, $d(a, b) = |\log a - \log b| \geq 0$ for all $a, b \in (0, \infty)$.

Also, $d(a, b) = 0$ if, and only if $\log a = \log b$ (since $d^{(1)}$ is a metric), which is true if, and only if, $a = b$ (since $x \mapsto \log x$ is strictly increasing on $(0, \infty)$.)

(M2) This is clear: $d(a, b) = |\log a - \log b| = |\log b - \log a| = d(b, a)$.

(M3) We have

$$\begin{aligned} d(a, c) &= |\log a - \log c| \\ &= |\log a - \log b + \log b - \log c| \\ &\leq |\log a - \log b| + |\log b - \log c| = d(a, b) + d(b, c) \end{aligned}$$

Having checked all three conditions we conclude that d is a metric on $\mathbb{R}$.

(Alternatively, we may use [Thm 1.2 \(HB, p74\)](#), noting that d is the pull-back of $d^{(1)}$ via the one-one function $f = \log : (0, \infty) \rightarrow \mathbb{R}$.)

Question 4

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Show there are infinitely many metrics possible on a set with two or more members.

How many metrics are possible on a singleton set?

Brief Answer

If X is a set with two or more members, and d_0 denotes the discrete metric on X , then for any $k \in (0, \infty)$, d_k is also a metric where $d_k(x, y) = kd_0(x, y)$.

This provides an infinite number of metrics on X .

If Y is a singleton set, say $Y = \{a\}$, then there is only one metric on Y , namely the map $d : Y \times Y \rightarrow \mathbb{R}$ given by $(a, a) \mapsto 0$.

Question 5

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This question looks at horizontal line segments in the plane.

A horizontal line segment in $(\mathbb{R}^2, d^{(2)})$ is determined by its mid-point (a, b) and a number $r \geq 0$ that governs its width, thus

$$H(a, b; r) = \{(x, y) \in \mathbb{R}^2 : a - r \leq x \leq a + r \text{ and } y = b\}.$$

Now let $\mathcal{H}$ be the set of all such line segments in $(\mathbb{R}^2, d^{(2)})$, that is:

$$\mathcal{H} = \{H(a, b; r) : (a, b) \in \mathbb{R}^2, r \geq 0\}.$$

Define a distance function D on $\mathcal{H}$ by:

$$D(H(a, b; r), H(a', b'; r')) = d^{(2)}((a, b), (a', b')) + |r - r'|$$

- (a) Sketch $H(-1, -1; 1)$, $H(1, 2; 3)$ and $H(0, 0; 0)$.
- (b) Calculate $D(H(-1, -1; 1), H(1, 2; 3))$.
- (c) Show that D defines a metric on $\mathcal{H}$.
- (d) Describe the unit sphere with centre $H(0, 0; 0)$.

Brief Answer - $H(a, b; r) = \{(x, y) \in \mathbb{R}^2: a - r \leq x \leq a + r \text{ and } y = b\}$.

(a)

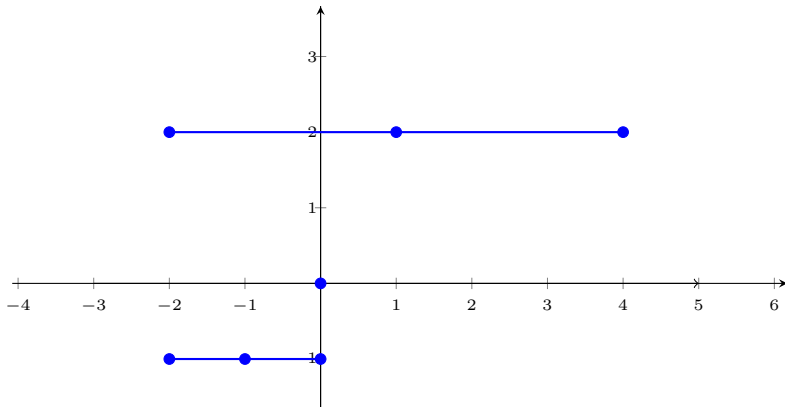
$$\begin{aligned} H(-1, -1; 1) &= \{(x, y) \in \mathbb{R}^2: -1 - 1 \leq x \leq -1 + 1 \text{ and } y = -1\} \\ &= \{(x, y) \in \mathbb{R}^2: -2 \leq x \leq 0 \text{ and } y = -1\} \\ &= [-2, 0] \times \{-1\}. \end{aligned}$$

(Here $[-2, 0] \times \{-1\}$ is a product as per [Defn 1.1, Book D p66](#).)

$$\begin{aligned} H(1, 2; 3) &= \{(x, y) \in \mathbb{R}^2: 1 - 3 \leq x \leq 1 + 3 \text{ and } y = 2\} \\ &= \{(x, y) \in \mathbb{R}^2: -2 \leq x \leq 4 \text{ and } y = 2\} \\ &= [-2, 4] \times \{2\}. \end{aligned}$$

$$\begin{aligned} H(0, 0; 0) &= \{(x, y) \in \mathbb{R}^2: 0 - 0 \leq x \leq 0 + 0 \text{ and } y = 0\} \\ &= \{(x, y) \in \mathbb{R}^2: x = 0 \text{ and } y = 0\} \\ &= \{(0, 0)\}. \end{aligned}$$

Brief Answer - $D(H(a, b; r), H(a', b'; r')) = d^{(2)}((a, b), (a', b')) + |r - r'|$
 (a)



(b)

$$\begin{aligned} D(H(-1, -1; 1), H(1, 2; 3)) &= d^{(2)}((-1, -1), (1, 2)) + |1 - 3| \\ &= \sqrt{13} + 2 \end{aligned}$$

Brief Answer $D(H(a, b; r), H(a', b'; r')) = d^{(2)}((a, b), (a', b')) + |r - r'|$

(c) To prove that D is a metric on $\mathcal{H}$, we will check the three conditions (M1)–(M3).

(M1) Clearly, by definition, $D(H(a, b; r), H(a', b'; r')) \geq 0$ for all $H(a, b; r), H(a', b'; r') \in \mathcal{H}$, and $D(H(a, b; r), H(a, b; r)) = 0$ for all $H(a, b; r) \in \mathcal{H}$.

Further if $H(a, b; r) \neq H(a', b'; r')$, then:

either $(a, b) \neq (a', b')$ and $d^{(2)}((a, b), (a', b')) > 0$,

or $r \neq r'$, in which case $|r - r'| > 0$.

Thus we see that (M1) is satisfied.

(M2) That this holds follows from the fact that both $d^{(2)}$ and $d^{(1)}$ are metrics.

Brief Answer

(c)(M3) Take any three segments $H(a, b; r)$, $H(a', b'; r')$ and $H(a'', b''; r'')$.

Then

$$\begin{aligned} & D(H(a, b; r), H(a'', b''; r'')) \\ &= d^{(2)}((a, b), (a'', b'')) + |r - r''| \\ &\leq d^{(2)}((a, b), (a', b')) + d^{(2)}((a', b'), (a'', b'')) + |r - r'| + |r' - r''| \\ &= \left(d^{(2)}((a, b), (a', b')) + |r - r'| \right) + \left(d^{(2)}((a', b'), (a'', b'')) + |r' - r''| \right) \\ &= D(H(a, b; r), H(a', b'; r')) + D(H(a', b'; r'), H(a'', b''; r'')) \end{aligned}$$

Thus we see that (M3) is satisfied.

Having checked all three axioms, we conclude that D defines a metric on $\mathcal{H}$.

Brief Answer

- (d) Recall that the unit sphere with centre $H(0, 0; 0)$ is given by $S_D(H(0, 0; 0), 1) = \{H(a, b; r) : D(H(0, 0; 0), H(a, b; r)) = 1\}$. Now

$$\begin{aligned} D(H(0, 0; 0), H(a, b; r)) &= d^{(2)}((0, 0), (a, b)) + |0 - r| \\ &= \sqrt{a^2 + b^2} + r = 1 \\ &\Leftrightarrow r = 1 - \sqrt{a^2 + b^2}. \end{aligned}$$

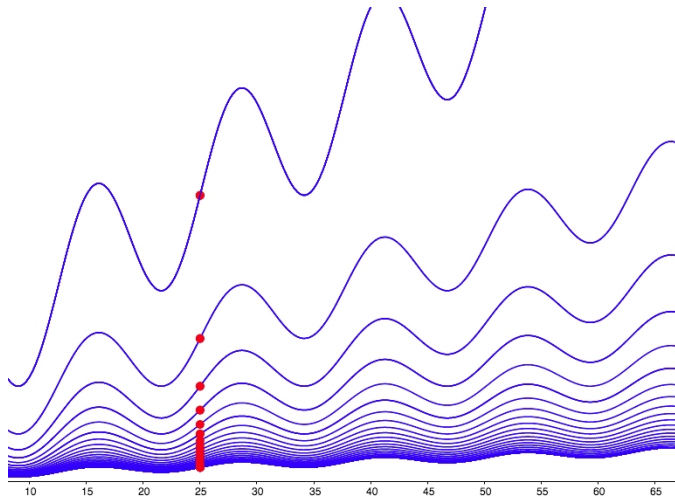
This means that to belong to that unit sphere, a segment $H(a, b; r)$ with mid-point (a, b) must have $\sqrt{a^2 + b^2} \leq 1$ and for any such (a, b) , the associated segment has $r = 1 - \sqrt{a^2 + b^2}$.

Bonus Points!

Lines segments in $S_D(H(0, 0; 0), 1)$ above are precisely those horizontal line segments that occur when horizontal lines intersect with a right triangle circumscribed in the (ordinary) unit circle.

- ▶ Pointwise Convergence is a concept that applied to sequences of (real) functions - (f_n) , where each f_n is a function with same domain $A \subseteq \mathbb{R}$.
- ▶ (*HB p78, Defn 4.1*)
A sequence (f_n) of functions $f_n : A \rightarrow \mathbb{R}$ converges pointwise to the function $f : A \rightarrow \mathbb{R}$ if, for each $x \in A$, the sequence of real numbers $(f_n(x))$ converges to $f(x)$.
- ▶ The pointwise limit of a sequence of continuous functions need not be continuous.

Pointwise Convergence - A summary

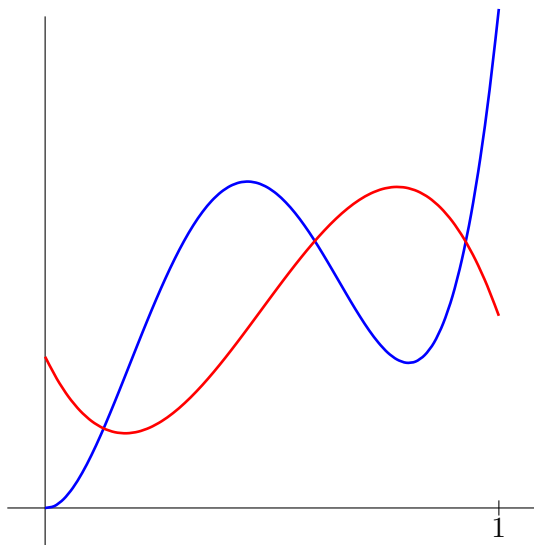


Uniform Convergence - A summary

- ▶ Uniform Convergence is a concept that applied to sequences of (real) functions - (f_n) , where each f_n is a function with same domain $A \subseteq \mathbb{R}$.
- ▶ (*HB p78, Defn 4.4*)
A sequence (f_n) of functions $f_n : A \rightarrow \mathbb{R}$ converges uniformly to f if:
 - (a) $f_n - f$ is bounded for each $n \in \mathbb{N}$.
 - (b) the sequence (M_n) defined by
$$M_n = \sup \{ |f_n(x) - f(x)| : x \in A \}$$
is a real null sequence.
- ▶ Uniform limit of a sequence of continuous functions will be continuous (*HB p78, Thm 4.7*).
- ▶ Closely related to uniform convergence is the max metric on $C[0,1]$:
$$d_{\max}(f, g) = \max \{ |f(x) - g(x)| : x \in [0, 1] \}$$

Uniform Convergence - A summary

$$d_{\max}(f, g) = \max\{|f(x) - g(x)| : x \in [0, 1]\}$$



Uniform Convergence - A summary

Theorem 5.8 *Uniform Convergence Theorem*

Let (f_n) be a sequence of functions in $C[0, 1]$.

If (f_n) converges uniformly on $[0, 1]$ to the function

$f : [0, 1] \rightarrow \mathbb{R}$, then f is continuous on $[0, 1]$ (and so is in $C[0, 1]$).

Furthermore $d_{\max}(f_n, f) \rightarrow 0$ as $n \rightarrow \infty$. Conversely, if

$f \in C[0, 1]$ and $d_{\max}(f_n, f) \rightarrow 0$ as $n \rightarrow \infty$, then (f_n) converges uniformly to f on $[0, 1]$.

Question 6

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Throughout this question we use the Euclidean metric for $\mathbb{R}$. For each $n \in \mathbb{N}$, define a function $f_n : [0, \infty) \rightarrow \mathbb{R}$ by

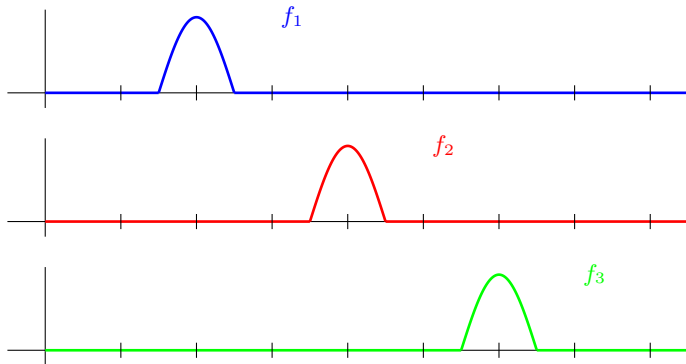
$$f_n(x) = \begin{cases} \cos(\pi x) & \text{if } 2n - \frac{1}{2} \leq x \leq 2n + \frac{1}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Sketch, by hand or otherwise, the graphs of f_1, f_2 and f_3 .
- (b) Determine whether the sequence (f_n) has a pointwise limit.
- (c) Determine whether the sequence (f_n) converges uniformly on $[0, \infty)$.

Brief Answer

$$f_n(x) = \begin{cases} \cos(\pi x) & \text{if } 2n - \frac{1}{2} \leq x \leq 2n + \frac{1}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Here are the graphs



Brief Answer

- (b) To show that the sequence (f_n) has a pointwise limit, we take $x \in [0, \infty)$ - fixed for the time being - and consider the sequence of numbers $(f_n(x))$.

Now, by the Archimedean Property of real numbers, we can find $N \in \mathbb{N}$ with $2N - 1/2 > x$.

Then for any $n > N$, $2n - 1/2 > x$ and so $f_n(x) = 0$. It follows that $(f_n(x))$ is convergent to 0.

And so we conclude that the sequence (f_n) has a pointwise limit function f , given by $f(x) = 0$ for all $x \in [0, \infty)$.

Brief Answer

- (c) To determine whether the sequence (f_n) has a uniform limit, we use *Lemma 4.5, HB p78*:

the only possible uniform limit is the pointwise limit f given above i.e. $f(x) = 0$ for all $x \in [0, \infty)$.

Notice first that for each $n \in \mathbb{N}$,

$|f_n(x) - f(x)| = |f_n(x) - 0| = |f_n(x)| \leq 1$ and this max occurs at $x = 2n$. Thus

► $f_n - f$ is bounded for each n , since

$$|(f_n - f)(x)| = |f_n(x)| \leq 1.$$

►

$$\begin{aligned} M_n &= \sup \{|f_n(x) - f(x)| : x \in [0, \infty)\} \\ &= 1 \end{aligned}$$

It follows that (M_n) is **not** a null sequence and so (f_n) does not converge uniformly on $[0, 1)$.

Brief Answer

Given $x \in [0, \infty)$ we want

$2N - 1/2 > x \Leftrightarrow 2N > x + 1/2 \Leftrightarrow N > x/2 + 1/4$, and such an N may be found by the Archimedean Property.

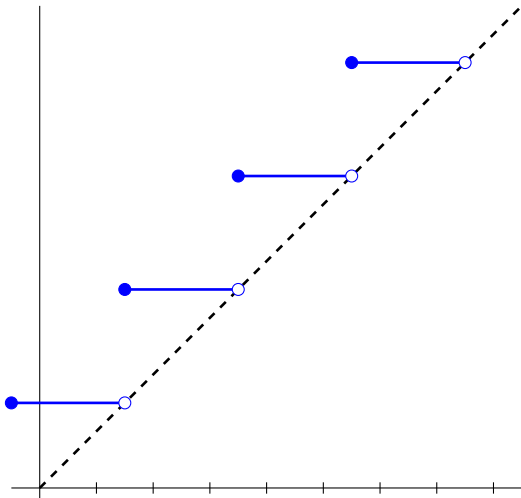
But we can actually give a formula for such an N . Let

$$\begin{aligned} N &= \text{int}(x/2 + 1/4) + 1 \\ &= \lfloor x/2 + 1/4 \rfloor + 1 \\ &= \lceil x/2 + 1/4 \rceil + 1 \end{aligned}$$

Then

$$\begin{aligned} 2N - 1/2 &= 2\text{int}(x/2 + 1/4) + 2 - 1/2 = 2\text{int}(x/2 + 1/4) + 3/2 \\ &> 2((x/2 + 1/4) - 1) + 3/2 \\ &= (x + 1/2 - 2) + 3/2 = x \end{aligned}$$

Brief Answer



Question 7

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Throughout this question we use the Euclidean metric for $\mathbb{R}$. For each $n \in \mathbb{N}$, define a function $f_n : [0, \infty) \rightarrow \mathbb{R}$ by

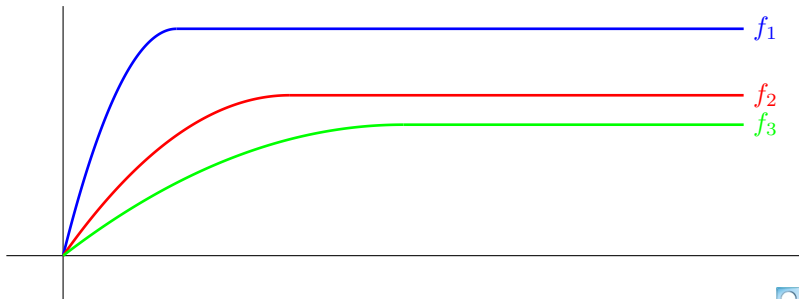
$$f_n(x) = \begin{cases} \frac{x(2n-x)}{n^2\sqrt{n}} & \text{if } 0 \leq x \leq n, \\ \frac{1}{\sqrt{n}} & \text{otherwise.} \end{cases}$$

- (a) Sketch, by hand or otherwise, the graphs of f_1, f_2 and f_3 .
- (b) Determine whether the sequence (f_n) has a pointwise limit.
- (c) Determine whether the sequence (f_n) converges uniformly on $[0, \infty)$.

Brief Answer

$$f_n(x) = \begin{cases} \frac{x(2n-x)}{n^2\sqrt{n}} & \text{if } 0 \leq x \leq n, \\ \frac{1}{\sqrt{n}} & \text{otherwise.} \end{cases}$$

(a) Here are the graphs



Brief Answer

- (b) To show that the sequence (f_n) has a pointwise limit, we take $x \in [0, \infty)$ and consider the sequence of numbers $(f_n(x))$. For $0 \leq x \leq n$, $f_n(x) = \frac{x(2n-x)}{n^2\sqrt{n}}$. Now elementary calculations show that $0 \leq x(2n-x) \leq n^2$ for $0 \leq x \leq 2n$. [Check!]

Thus, for $0 \leq x \leq n$ we have $0 \leq f_n(x) \leq \frac{n^2}{n^2\sqrt{n}} \leq \frac{1}{\sqrt{n}}$.

But for $x > n$, $f_n(x) = \frac{1}{\sqrt{n}}$ and so we have:

$$|f_n(x)| \leq \frac{1}{\sqrt{n}} \text{ for all } n \in \mathbb{N}, x \in \mathbb{R}.$$

Brief Answer

$$(b) \quad |f_n(x)| \leq \frac{1}{\sqrt{n}} \text{ for all } n \in \mathbb{N}, x \in \mathbb{R}.$$

Thus given any $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon^2}$ (so that $\frac{1}{\sqrt{N}} < \varepsilon$).

Hence for $n > N$

$$|f_n(x)| \leq \frac{1}{\sqrt{n}} < \frac{1}{\sqrt{N}} < \varepsilon.$$

Thus $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$.

We conclude that the sequence (f_n) has a pointwise limit function f , given by $f(x) = 0$ for all $x \in [0, \infty)$.

Brief Answer

- (c) To determine whether the sequence (f_n) has a uniform limit, we use *Lemma 4.5, HB p78*: the only possible uniform limit is the function f given above.

Notice first that for each $n \in \mathbb{N}$, the maximum value of $|f_n(x) - f(x)| = \frac{1}{\sqrt{n}}$ and occurs at $x = n$. Thus

- ▶ $f_n - f$ is bounded for each n , since

$$|(f_n - f)(x)| = |f_n(x)| \leq \frac{1}{\sqrt{n}}.$$



$$\begin{aligned} M_n &= \sup \{|f_n(x) - f(x)| : x \in [0, \infty)\} \\ &= \frac{1}{\sqrt{n}} \end{aligned}$$

It follows that $M_n \rightarrow 0$ as $n \rightarrow \infty$.

Thus (f_n) has a uniform limit f .

Question 8

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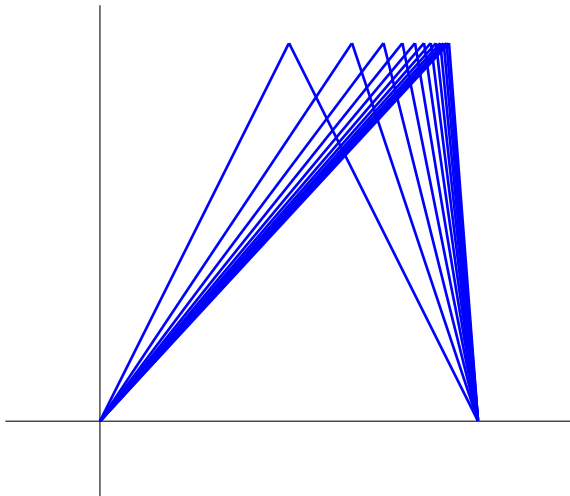
For $n = 1, 2, 3, \dots$ define $f_n : [0, 1] \rightarrow [0, 1]$ by

$$f_n = \begin{cases} \frac{(n+1)x}{n} & \text{if } 0 \leq x \leq \frac{n}{n+1}, \\ (n+1)(1-x) & \text{if } \frac{n}{n+1} < x \leq 1. \end{cases}$$

Show that (f_n) has a pointwise limit on $[0, 1]$ and decide whether (f_n) is $d_{\max}$ convergent to the pointwise limit.

Brief Answer

Some sketches.



Brief Answer

First of all, choose x , with $0 \leq x < 1$ then choose $N \in \mathbb{N}$ with $\frac{N}{N+1} > x$.

Then for $n > N$, we have $f_n(x) = \frac{(n+1)x}{n} \rightarrow x$ as $n \rightarrow \infty$.

Also, for $x = 1$, $f_n(x) = f_n(1) = 0$ for all $n \in \mathbb{N}$.

It follows that (f_n) has a pointwise limit f on $[0, 1]$ given by

$$f = \begin{cases} x & \text{if } 0 \leq x < 1, \\ 0 & \text{if } x = 1. \end{cases}$$

As this function is not continuous on $[0, 1]$ it follows, in the light of [Thm 5.8, HB, p79](#) and the fact that each f_n is continuous, that (f_n) is not $d_{\max}$ convergent to this pointwise limit.

Question 9

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Show that the function $F : C[0, 1] \rightarrow C[0, 1]$ is $(d_{\max}, d_{\max})$ -continuous on $C[0, 1]$ where

$$F(f)(t) = \sin(t) + \int_0^t \cos(x)f(x) dx$$

for each $f \in C[0, 1]$ and each $t \in [0, 1]$.

Brief Answer

We show that F is $(d_{\max}, d_{\max})$ -Lipschitz which will prove the result by *Thm. 3.2, HB p76*. So take any two functions $f, g \in C[0, 1]$:

$$\begin{aligned} |F(f)(t) - F(g)(t)| &= \left| \left(\sin(t) + \int_0^t \cos(x) f(x) dx \right) - \left(\sin(t) + \int_0^t \cos(x) g(x) dx \right) \right| \\ &= \left| \left(\int_0^t \cos(x) f(x) dx - \int_0^t \cos(x) g(x) dx \right) \right| \\ &= \left| \int_0^t \cos(x) (f(x) - g(x)) dx \right| \\ &\leq \int_0^t |\cos(x) (f(x) - g(x))| dx \quad (\text{See } \textit{HB, p12 - final line.}) \\ &= \int_0^t |\cos(x)| |f(x) - g(x)| dx \\ &\leq \int_0^t |f(x) - g(x)| dx \leq \int_0^1 d_{\max}(f, g) dx = d_{\max}(f, g). \end{aligned}$$

Brief Answer

The inequalities above gave us:

$$|F(f)(t) - F(g)(t)| \leq d_{\max}(f, g)$$

From this it follows that:

$\max \{|F(f)(t) - F(g)(t)| : t \in [0, 1]\} \leq d_{\max}(f, g)$ and so
 $d_{\max}(F(f), F(g)) \leq d_{\max}(f, g)$.

Thus F is $(d_{\max}, d_{\max})$ -Lipschitz (with constant 1) and so is $(d_{\max}, d_{\max})$ -continuous on $C[0, 1]$.

A very concise answer exploits *HB, p79 - Examples*:

$$F = F_{\sin} \circ H \circ G_{\cos}.$$

Brief Answer

Question 10

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For each $f \in C[0, 1]$ define $F : C[0, 1] \rightarrow \mathbb{R}$ by

$$F(f) = 2f(0) + 3 \int_0^1 sf(1-s)ds$$

Show that F is $(d_{\max}, d^{(1)})$ -continuous on $C[0, 1]$.

Brief Answer

Some examples,

f	$F(f) = 2f(0) + 3 \int_0^1 s f(1-s) ds$
$f(x) = 1$	$F(f) = 2 + 3 \int_0^1 s ds = 7/2$
$f(x) = x$	$F(f) = 0 + 3 \int_0^1 s(1-s) ds = 1/2$
$f(x) = x^2$	$F(f) = 0 + 3 \int_0^1 s(1-s)^2 ds = 1/4$
$f(x) = e^x$	$F(f) = 2 + 3 \int_0^1 s e^{(1-s)} ds = 3e - 4$

Brief Answer

We show that F is $(d_{\max}, d^{(1)})$ -Lipschitz which will prove the result by [Thm. 3.2](#), [HB p76](#). To this end, we take any two functions $f, g \in C[0, 1]$:

$$\begin{aligned} d^{(1)}(F(f), F(g)) &= |F(f) - F(g)| \\ &= \left| \left(2f(0) + 3 \int_0^1 s f(1-s) ds \right) - \left(2g(0) + 3 \int_0^1 s g(1-s) ds \right) \right| \\ &= \left| 2(f(0) - g(0)) + 3 \int_0^1 s (f(1-s) - g(1-s)) ds \right| \\ &\leq 2|f(0) - g(0)| + 3 \left| \int_0^1 s (f(1-s) - g(1-s)) ds \right| \\ &\leq 2d_{\max}(f, g) + 3 \int_0^1 |s| |f(1-s) - g(1-s)| ds \\ &\leq 2d_{\max}(f, g) + 3 \int_0^1 |f(1-s) - g(1-s)| ds \\ &\leq 2d_{\max}(f, g) + 3 \int_0^1 d_{\max}(f, g) ds = 5d_{\max}(f, g). \end{aligned}$$

Thus F is $(d_{\max}, d^{(1)})$ -Lipschitz (with constant 5) and so is $(d_{\max}, d^{(1)})$ -continuous.

- ▶ A - in (X, d) - is d -closed if for each d -convergent sequence with terms in A , the limit of the sequence is also in A . ([HB, p80](#))
- ▶ A point $a \in X$ is a d -closure point of A - in (X, d) - if for every $r > 0$, $B_d(a, r)$ meets A (that is intersects with A). ([Defn 3.6, HB, p84](#)).
- ▶ A point $a \in X$ is a d -closure point of A - in (X, d) - if there is a sequence (a_n) in A that converges to a . ([HB, p84](#)).
- ▶ The d -closure of A - $\text{Cl}_{(X,d)}(A)$ - is the set of all d -closure points of A ([HB, p84](#)).
- ▶ A - in (X, d) - is d -closed if, and only if, $A = \text{Cl}_{(X,d)}(A)$.
- ▶ It is always true that $A \subseteq \text{Cl}_{(X,d)}(A)$. ([HB, p84](#)).
- ▶ For the discrete metric, *all* sets are closed.

Open Sets - A summary

- ▶ A - in (X, d) - is d -open if, and only if, for each $a \in A$ there is a d -open ball with centre a and $r > 0$ that lies wholly in A . ([HB, p82](#)).
- ▶ A point $a \in X$ is a d -interior point of A - in (X, d) - if **there exists** a d -open ball $B_d(a, r)$ with centre a and $r > 0$ that lies wholly in A . ([HB, p85](#)).
- ▶ The d -interior of A - $\text{Int}_{(X,d)}(A)$ - is the set of all d -interior points of A ([HB, p85](#)).
- ▶ A - in (X, d) - is d -open if, and only if, $A = \text{Int}_{(X,d)}(A)$.
- ▶ It is always true that $\text{Int}_{(X,d)}(A) \subseteq A$. ([HB, p85](#)).
- ▶ A is d -open if, and only if, the complement of A , $A^c = X - A$ is d -closed ([HB, p80](#)).
- ▶ For the discrete metric, *all* sets are open.

Boundaries - A summary

- ▶ A point $x \in X$ is a d -boundary of a set A if, for each $r > 0$, $B(x, r)$ meets both A and A^c ([HB, p86](#)).
- ▶ The d -boundary of A - in (X, d) - is
$$\text{Bd}_{(X,d)}(A) = \text{Cl}_{(X,d)}(A) - \text{Int}_{(X,d)}(A) \quad (\text{HB, p85}).$$

“ boundary = closure - interior ”

Question 11

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Let

$$A = \{(a, b) \in \mathbb{R}^2 : 2 < |a| + |b| \leq 4\} \quad B = \{(a, b) \in \mathbb{R}^2 : b < |a|\}$$

and $C = A - B$.

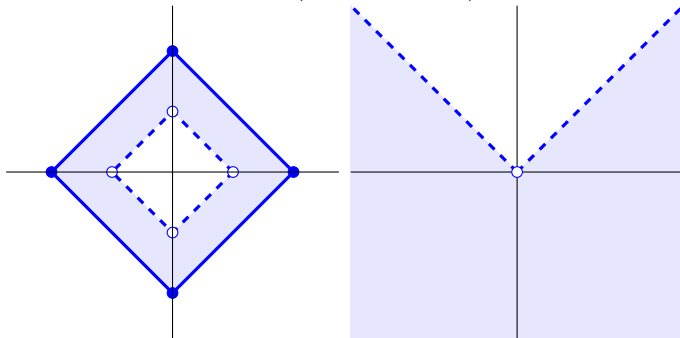
- (a) Sketch A, B and C .
- (b) Is $(1, 1)$ a closure point of C with respect to the Euclidean metric $d^{(2)}$?
- (c) Write down in set notation the closure, interior and boundary of C with respect to $d^{(2)}$.
- (d) Write down, in set notation, the closure, interior and boundary of C for the ‘mixed’ metric d on $\mathbb{R}^2$ given by

$$d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + d_0(x_2, y_2)$$

.

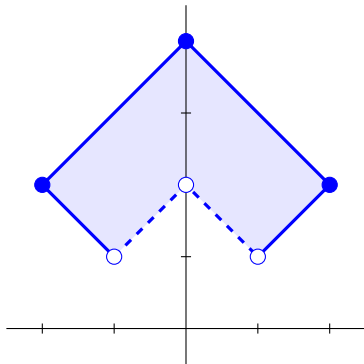
Brief Answer

A, B are shown below (A on the left).



Brief Answer

And here is C .



Brief Answer

(b) $(1, 1)$ is a $d^{(2)}$ -closure point of C .

Perhaps the simplest way to see this is to note that $(1, 1 + \frac{1}{n}) \in C$ for all $n \in \mathbb{N}$ and

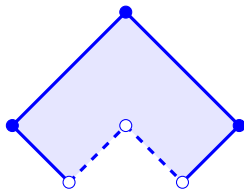
$$d^{(2)}\left(\left(1, 1 + \frac{1}{n}\right), (1, 1)\right) = \frac{1}{n} \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Thus

$$\left(1, 1 + \frac{1}{n}\right) \xrightarrow{d^{(2)}} (1, 1), \text{ as } n \rightarrow \infty.$$

as required.

Brief Answer



(c) Closure, interior, boundary for $d^{(2)}$

$$\text{Cl}_{(\mathbb{R}^2, d^{(2)})}(C) =$$

$$\{(a, b) \in \mathbb{R}^2 : 2 - |a| \leq b \leq 4 - |a| \text{ and } b \geq |a|\}$$

Brief Answer

$$(c) \operatorname{Int}_{(\mathbb{R}^2, d^{(2)})}(C) = \{(a, b) \in \mathbb{R}^2 : 2 - |a| < b < 4 - |a| \text{ and } b > |a|\}$$

$$\begin{aligned} \operatorname{Bd}_{(\mathbb{R}^2, d^{(2)})}(C) = & \{(a, b) \in \mathbb{R}^2 : b = 2 - |a| \text{ and } |a| \leq 1\} \\ & \cup \{(a, b) \in \mathbb{R}^2 : b = 4 - |a| \text{ and } |a| \leq 2\} \\ & \cup \{(a, b) \in \mathbb{R}^2 : b = |a| \text{ and } 1 \leq |a| \leq 2\} \end{aligned}$$

Brief Answer

(d) For the ‘mixed’ metric,

$$d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + d_0(x_2, y_2)$$

$$\text{Cl}_{(\mathbb{R}^2, d)}(C) = \text{Cl}_{(\mathbb{R}^2, d^{(2)})}(C) - \{(1, 1), (-1, 1)\}$$

$$\text{Int}_{(\mathbb{R}^2, d)}(C) = \text{Int}_{(\mathbb{R}^2, d^{(2)})}(C)$$

$$\text{Bd}_{(\mathbb{R}^2, d)}(C) = \text{Bd}_{(\mathbb{R}^2, d^{(2)})}(C) - \{(1, 1), (-1, 1)\}$$

Question 12

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- (a) Can you find a single metric space (A, d) where:
- (i) The interval $(0, 1]$ is both d -open and d -closed,
 - (ii) The interval $[2, 3)$ is d -closed but not d -open,
 - (iii) The set $\{4\}$ is d -closed but not d -open,
 - (iv) The set $\{5\}$ is d -closed and d -open.
- (b) Is there a metric space (A, d) where, for some $a \in A$, $\{a\}$ is open but not closed?

Brief Answer

- (a) One example is (A, d) where A is the subset of $\mathbb{R}$ given by:
 $A = (0, 1] \cup (\frac{3}{2}, 3) \cup (\frac{7}{2}, \frac{9}{2}) \cup \{5\}$, and $d = d_A^{(1)}$ the metric induced from $d^{(1)}$ on $\mathbb{R}$.
- (i) Then $(0, 1]$ is d_A -open as $(0, 1] = A \cap (0, \frac{3}{2})$ and $(0, \frac{3}{2})$ is $d^{(1)}$ -open. And $(0, 1]$ is d_A -closed as $(0, 1] = A \cap [0, 1]$ and $[0, 1]$ is $d^{(1)}$ -closed.
- (ii) The interval $[2, 3)$ is d_A -closed as $[2, 3) = A \cap [2, 3]$ and $[2, 3]$ is $d^{(1)}$ -closed.
The interval $[2, 3)$ is not d_A -open as its complement (in A) is not d_A -closed. The complement includes $(\frac{3}{2}, 2)$ which contains sequences that are d_A -convergent to 2 (in A), which is not in that complement. $[2, 3) = A \cap [2, 3]$ and $[2, 3]$ is $d^{(1)}$ -closed.
- (iii) Argue as in (ii) to show that $\{4\}$ is d_A -closed but not d_A -open,
- (iv) Argue as in (i) to show that $\{5\}$ is d_A -closed and d_A -open.
- (b) No. By [Lemma 1.2, HB p80](#) all singleton sets are closed, so there is no metric space (A, d) where $\{a\}$ is open but not closed.

Question 13

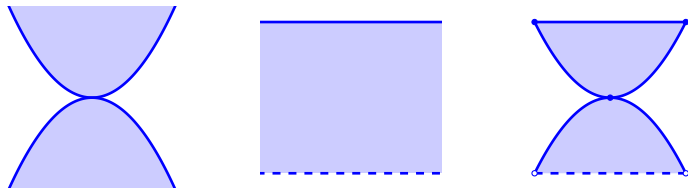
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$A = \{(a, b) \in \mathbb{R}^2 : |b| \geq a^2\}$, $B = \{(a, b) \in \mathbb{R}^2 : -1 < b \leq 1\}$,
and $C = A \cap B$.

- (a) Sketch on separate diagrams each of A , B and C .
- (b) Determine whether $(0, 0)$ is a $d^{(2)}$ -boundary point of C .
- (c) Write down, in set notation, the closure, interior and boundary of C for:
 - (i) the Euclidean metric $d^{(2)}$,
 - (ii) the ‘mixed’ metric d on $\mathbb{R}^2$ given by
$$d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + d_0(x_2, y_2),$$
 - (iii) the discrete metric d_0 .

Brief Answer

(a) Here are the sketches (A left; C right).



(b) The point $(0,0)$ is a $d^{(2)}$ -boundary point of C . We use *Defn 5.3, HB p86*.

Take any $r > 0$ and consider the $d^{(2)}$ -ball $B_{d^{(2)}}((0,0), r)$.

Certainly, this ball meets C , since $(0,0) \in C$.

Also, $(r/2, 0)$ lies both in C^c and $B_{d^{(2)}}((0,0), r)$, showing that $B_{d^{(2)}}((0,0), r)$ meets C^c as well.

It follows that $(0,0)$ is a $d^{(2)}$ -boundary point of C .

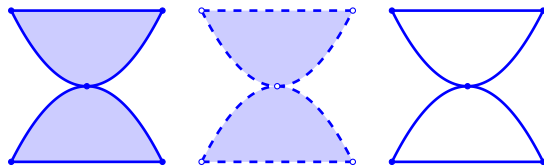
Brief Answer

(c) (i) For the metric $d^{(2)}$,

$$\text{Cl}(C) = \{(a, b) \in \mathbb{R}^2 : |b| \geq a^2 \text{ and } |b| \leq 1\},$$

$$\text{Int}(C) = \{(a, b) \in \mathbb{R}^2 : |b| > a^2 \text{ and } |b| < 1\},$$

$$\begin{aligned} \text{Bd}(C) = & \{(a, b) \in \mathbb{R}^2 : |b| = a^2 \text{ and } |a| \leq 1\} \\ & \cup \{(a, 1) \in \mathbb{R}^2 : |a| \leq 1\} \cup \{(a, -1) \in \mathbb{R}^2 : |a| \leq 1\}. \end{aligned}$$



Brief Answer

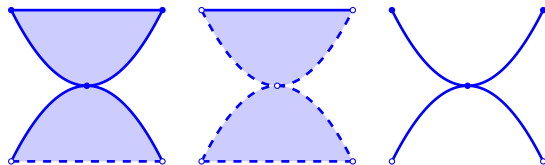
(c) (ii) For the 'mixed' metric d ,

$$\text{Cl}(C) = \{(a, b) \in \mathbb{R}^2 : |b| \geq a^2 \text{ and } -1 < b \leq 1\} (= C),$$

$$\text{Int}(C) =$$

$$\{(a, b) \in \mathbb{R}^2 : |b| > a^2 \text{ and } |b| < 1\} \cup \{(a, 1) \in \mathbb{R}^2 : |a| < 1\},$$

$$\text{Bd}(C) = \{(a, b) \in \mathbb{R}^2 : |b| = a^2 \text{ and } -1 < b \leq 1\}.$$



(iii) For the discrete metric d_0 ,

$$\text{Cl}(C) = C, \quad \text{Int}(C) = C \quad \text{Bd}(C) = \emptyset.$$

NB For the discrete metric, all sets are open; all sets are closed. (HB p89)

Question 14

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B, C, D and E are subsets of $\mathbb{R}^2$ given by:

$$B = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 < 2\}, \quad C = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > |x_1|\},$$

$$D = \{(x_1, x_2) \in \mathbb{R}^2 : 2 < x_2 \leq 4\},$$

$$E = \{(x_1, x_2) \in \mathbb{R}^2 : -1 < x_1 \leq 1\}$$

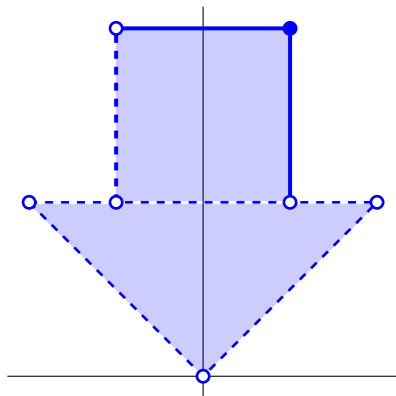
and $A = (B \cap C) \cup (D \cap E)$.

- (a) Sketch the set A , following the conventions used in the module text for such sketches.
- (b) Determine the d -interior, d -closure and d -boundary of C when d is:
 - (i) the Euclidean metric $d^{(2)}$,
 - (ii) the ‘mixed’ metric d given by
$$d((x_1, x_2), (y_1, y_2)) = d_0(x_1, y_1) + |x_2 - y_2|,$$
where (as usual) d_0 is the discrete metric on $\mathbb{R}$.

Brief Answer

$B = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 < 2\}$, $C = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > |x_1|\}$,
 $D = \{(x_1, x_2) \in \mathbb{R}^2 : 2 < x_2 \leq 4\}$, $E = \{(x_1, x_2) \in \mathbb{R}^2 : -1 < x_1 \leq 1\}$
and $A = (B \cap C) \cup (D \cap E)$.

(a) Here is the sketch of A .



Brief Answer

(b) (i) the $d^{(2)}$ -interior is: $(B \cap C) \cup (D' \cap E')$, where

$$D' = \{(x_1, x_2) \in \mathbb{R}^2 : 2 < x_2 < 4\} \text{ and}$$

$$E' = \{(x_1, x_2) \in \mathbb{R}^2 : |x_1| < 1\},$$

the $d^{(2)}$ -closure is: $(B' \cap C') \cup (D'' \cap E'')$, where

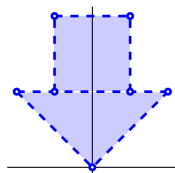
$$B' = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq 2\}, C' = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \geq |x_1|\},$$

$$D'' = \{(x_1, x_2) \in \mathbb{R}^2 : 2 \leq x_2 \leq 4\} \text{ and}$$

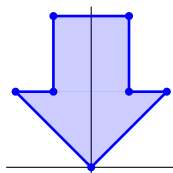
$$E'' = \{(x_1, x_2) \in \mathbb{R}^2 : |x_1| \leq 1\}.$$

The $d^{(2)}$ -boundary is the set difference between the $d^{(2)}$ -closure and the $d^{(2)}$ -interior:

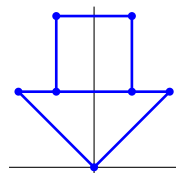
$$\{(x_1, x_2) \in \mathbb{R}^2 : x_2 = |x_1| \text{ and } |x_1| \leq 2\} \cup \{(x_1, 2) \in \mathbb{R}^2 : |x_1| \leq 2\} \\ \cup \{(x_1, x_2) \in \mathbb{R}^2 : |x_1| = 1 \text{ and } 2 \leq x_2 \leq 4\} \cup \{(x_1, 4) \in \mathbb{R}^2 : |x_1| \leq 1\}.$$



$d^{(2)}$ -interior



$d^{(2)}$ -closure



$d^{(2)}$ -boundary

Brief Answer

(b) (ii) For the 'mixed' metric d given by

$$d((x_1, x_2), (y_1, y_2)) = d_0(x_1, y_1) + |x_2 - y_2|:$$

the d -interior is: $(B \cap C) \cup (D' \cap E)$, where

$$D' = \{(x_1, x_2) \in \mathbb{R}^2 : 2 < x_2 < 4\},$$

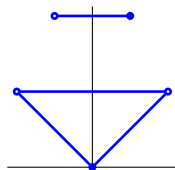
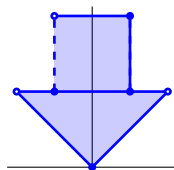
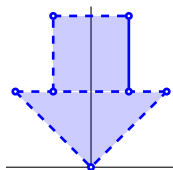
the d -closure is: $(B \cap C') \cup (D \cap E'')$.

The d -boundary is the set difference between the d -closure and the d -interior:

$$\{(x_1, x_2) \in \mathbb{R}^2 : -1 < x_1 \leq 1 \text{ and } x_2 = 4\} \cup$$

$$\{(x_1, x_2) \in \mathbb{R}^2 : -2 < x_1 < 2 \text{ and } x_2 = 2\} \cup$$

$$\{(x_1, x_2) \in \mathbb{R}^2 : x_2 = |x_1| \text{ and } |x_1| < 2\}$$



Question 15

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Throughout, for convenience, we use d for the Euclidean metric on $\mathbb{R}$.

Further, we will use $\mathbb{P}$ to denote the set of prime numbers in $\mathbb{N}$.

Let $S = \{x \in \mathbb{R} : px^2 = (n-1)^2 \text{ for some } p \in \mathbb{P}, n \in \mathbb{N}\}$.

- (a) Show that S is countable.
- (b)
 - (i) Show that for each $y \in \mathbb{R}$ and each $r > 0$ there is $x \in S$ with $|x - y| < r$.
 - (ii) Hence, or otherwise, show that S is d -dense in $\mathbb{R}$.

Brief Answer

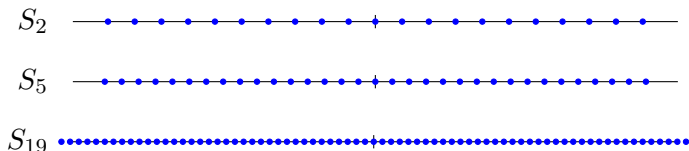
We begin by introducing, for convenience, two auxiliary sets:

For given $p \in \mathbb{P}$ and $n \in \mathbb{N}$ define $S_{p,n}$ by

$$S_{p,n} = \{x \in \mathbb{R} : px^2 = (n-1)^2\}$$

and for $p \in \mathbb{P}$ define S_p by

$$S_p = \{x \in \mathbb{R} : px^2 = (n-1)^2 \text{ for } n \in \mathbb{N}\}.$$



Brief Answer

- (a) Here is the most concise answer: each $S_{p,n}$ is finite and so countable; $\mathbb{P} \times \mathbb{N}$ is a subset of $\mathbb{Q} \times \mathbb{Q}$ and so is countable. Thus

$$S = \bigcup_{(p,n) \in \mathbb{P} \times \mathbb{N}} S_{p,n}$$

expresses S as the countable union of countable sets and so is countable. Expanding this slightly, we have

$S_{p,n}$ is finite and so countable (See *Remark 1, BkD, p228*)

$\bigcup_{n=1}^{\infty} S_{p,n}$ is a countable union of countable sets and so is countable (*HB p 87 or Bk D p230*)

$\bigcup_{p \in \mathbb{P}} \bigcup_{n=1}^{\infty} S_{p,n}$ is a countable union of countable sets and so is countable (*HB p 87 or Book D p230*)

So S itself is countable.

Brief Answer

- (b) (i) We begin by noting that the set S_p may be written as $S_p = \{x \in \mathbb{R} : x = k/\sqrt{p} \text{ for } k \in \mathbb{Z}\}$ and that the distance between successive members S_p is

$$\left| \frac{k+1}{\sqrt{p}} - \frac{k}{\sqrt{p}} \right| = \frac{1}{\sqrt{p}}$$

Now for any $y \in \mathbb{R}$ and $r > 0$ we choose $p \in \mathbb{P}$ with $p > 1/r^2$, so that $\frac{1}{\sqrt{p}} < r$.

It follows that there is $x \in S_p$ (and so in S) with $x \in (y - r, y + r)$ that is with $|x - y| < r$.



(Informally, the numbers in S_p - shown in red above - are spaced sufficiently closely so that at least one of them falls in the blue interval shown.)

Brief Answer

- (b) (ii) The result of the previous part may be rephrased as follows: given any $y \in \mathbb{R}$, every d -open ball with centre y contains a member of the set S .

Thus every number in $y \in \mathbb{R}$ is a d -closure point of S .

On the other hand, as $S \subseteq \mathbb{R}$, we know that the d -closure of S , $\text{Cl}_{(\mathbb{R},d)}(S)$, is a subset of $\mathbb{R}$ (by the 3rd “Basic Property” at the top of (HB p 84)).

Hence, $\text{Cl}_{(\mathbb{R},d)}(S) = \mathbb{R}$ and we conclude that S is d -dense in $\mathbb{R}$ (Defn 6.1, HB p 86).

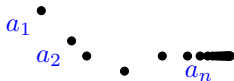
Appendix - mixed metrics

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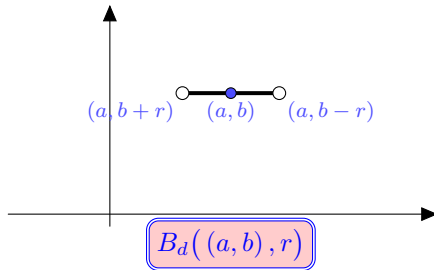
For the mixed metric $d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + d_0(x_2, y_2)$

i.e. Euclidean in **first** coordinate, we have:

Any d -convergent sequence must eventually be constant in the second coordinate.



Small d -open balls look like this



Some comments on mixed metrics

For the mixed metric $d((x_1, x_2), (y_1, y_2)) = d_0(x_1, y_1) + |x_2 - y_2|$
i.e. Euclidean in ***second*** coordinate, we have:

Any d -convergent sequence must eventually be constant in the first coordinate.



Small d -open balls look like this

